

“We Bring Brain-Like Intelligence Directly To Sensors For Ultra-Low-Power AI Processing”

Sensors generate more data than the cloud can handle. Innatera's neuromorphic chips shift AI to the edge, enabling always-on, ultra-low-power intelligence at the source. In an exclusive EFY conversation, Sumeet Kumar of Innatera discusses with EFY's Ashwini Kumar Sinha, Nidhi Agarwal, and Saba Aafreen how hybrid spiking and conventional neural networks are defining the next wave of smart edge AI.



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How are neuromorphic principles implemented at the hardware level?

In the case of the Pulsar chip, neuromorphic principles are implemented by mimicking the brain's hardware and software structure. The chips use processing elements that function as silicon neurons and synapses. These recreate integrate-and-fire spiking neuron behaviour in silicon, enabling time-series integration and fine-grained temporal spike generation. This approach allows spiking neural networks (SNNs) to be reproduced in silicon with full fidelity. A significant part of the engineering effort focuses on making these neurons and synapses robust, high-performing, energy-efficient, and manufacturable, while organising them into a programmable fabric.

What are the main compute elements in the neuromorphic microcontrollers?

The Pulsar chip integrates three major compute fabrics. The first is a spiking neural network accelerator built from processing elements that mimic biological neurons and synapses. These silicon neurons operate in parallel and consume

extremely small amounts of energy per operation.

The second element is a conventional CNN accelerator that enables developers to efficiently run traditional deep learning models as needed. The third component is a RISC-V CPU subsystem with standard sensor interfaces. This CPU handles tasks such as sensor data acquisition, control logic, and actuation. Together, these elements form a complete system-on-chip that provides all the processing resources required for sensing applications while maintaining extremely low power consumption.

What is SNN processing and why is it important in neuromorphic systems?

In conventional neural networks, data is processed as continuous digital values, and every node consumes energy regardless of whether the computation is meaningful. Spiking neural networks are fundamentally different. They are event-driven. Sensor data is encoded as simple voltage spikes, essentially single-bit events, where information is carried in the timing or frequency of the spikes. Computation occurs

only when something meaningful happens. These networks inherently understand time, enabling them to be smaller and far more energy-efficient than traditional neural networks. This temporal, event-driven nature closely mirrors how the biological brain processes information, making SNNs particularly effective for edge and sensor-based applications.

How do you train the SNNs?

From a developer's perspective, training SNNs is similar to training conventional neural networks. Innatera has developed a software framework called Talamo that uses PyTorch as its front end. Developers define, train, and optimise their models using standard PyTorch workflows. Once the model is ready, the company's compilation framework automatically maps it onto the hardware. Engineers do not need to understand the chip's internal details, making deployment straightforward and familiar for teams already working with modern AI tools.

Why do developers still prefer CNNs for image and video processing?

Traditional image and video data